

Visvesvaraya Technological University

Belagavi, Karnataka

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ

ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A Project Synopsis

ON

"Product Originality Detector using Blockchain"

Submitted to Visvesvaraya Technological University in partial fulfilment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

Submitted by

Project Batch No: C5

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I. Problem Description

1. Problems identified

The product originality detector faces several challenges that hinder its effectiveness. One major issue is the complexity of counterfeit detection, as counterfeiters continuously develop advanced techniques to mimic genuine products. Additionally, the system may struggle with scalability, especially when dealing with a vast range of products across different industries. The reliability of authentication methods, such as QR (Quick Response) codes or RFID (Radio Frequency Identification) tags, can also be compromised if counterfeiters find ways to replicate them. Furthermore, user adoption remains a challenge since consumers and businesses may not always be aware of or willing to use the system. The cost of implementing and maintaining such a system can be high, making it difficult for small businesses to afford. Another problem is data security, as counterfeit identification systems often involve the storage and processing of sensitive product information. The system's effectiveness also depends on real-time updates, which can be difficult to achieve in large supply chains. Additionally, regulatory and legal challenges may arise, especially when enforcing anti-counterfeit laws across different jurisdictions. Finally, counterfeiters may exploit loopholes in the system, requiring continuous improvements and updates to stay ahead of fraudulent activities.

2. Existing solutions

Several existing solutions have been developed to combat counterfeit products using advanced technology and authentication methods. One widely used solution is QR code-based verification, where consumers can scan a unique code on the product packaging to verify its authenticity. RFID and NFC (Near Field Communication) tags are also implemented in high-value products to enable real-time tracking and authentication. Blockchain technology has emerged as a robust solution, providing a decentralized and tamper-proof ledger for supply chain transparency. Artificial intelligence and machine learning are being used to analyse patterns and detect anomalies in product authentication. Holograms and security labels with unique identifiers are also used to make duplication difficult for counterfeiters. Governments and organizations are increasingly adopting centralized databases that allow businesses and consumers to cross-check product information. Mobile applications integrated with authentication systems provide instant verification, making it easier for consumers to identify fake products. Furthermore,

companies are partnering with law enforcement agencies to strengthen anti-counterfeiting regulations and take legal action against counterfeiters. Despite these solutions, ongoing advancements and improvements are necessary to keep pace with evolving counterfeit techniques.

3. Proposed solution and its advantages

The product originality detector offers a comprehensive and innovative approach to tackling counterfeit goods using cutting-edge technology. By integrating blockchain technology, the system ensures a tamper-proof and transparent record of product authenticity, making it nearly impossible for counterfeiters to manipulate data. The system also employs advanced QR codes and RFID/NFC tags, allowing consumers and businesses to instantly verify products using provided solution. Artificial intelligence enhances detection by analysing purchasing patterns and identifying potential counterfeit activities in real-time. One of the key advantages of the solution is its user-friendly interface, enabling seamless adoption by both manufacturers and consumers. Further, the system provides real-time tracking and authentication across the supply chain, ensuring complete transparency from production to the end user. The proposed solution is cost-effective and scalable, making it accessible to businesses of all sizes without the need for extensive infrastructure investment. Moreover, by collaborating with regulatory bodies and law enforcement agencies, system enhances legal enforcement against counterfeiters. Overall, product originality detector offers a highly secure, efficient, and scalable solution that significantly reduces the risks posed by counterfeit goods.

II. Mapping to the Program Outcomes (Pos)

PO#	Level	Justification
PO3	3	Develop a robust product originality detector that integrates advanced authentication techniques to ensure product integrity. The solution prioritizes public health and safety while considering cultural, societal, and environmental impacts, enhancing consumer trust and regulatory compliance.

PO5	2	This enables the use of advanced authentication technologies, such as AI, blockchain, and RFID, for counterfeit detection.
PO6	1	This ensures the system addresses legal regulations, consumer safety, and ethical responsibilities in product authentication.
PO7	2	This promotes sustainability by reducing the environmental impact of counterfeit goods and ensuring eco-friendly verification methods.

Expected Outcomes

The expected outcomes of the product originality detector include enhanced product authenticity, reduced counterfeit goods, improved consumer trust, increased brand protection, better supply chain transparency, real-time verification, stronger legal enforcement, cost savings for businesses, and higher adoption of secure authentication methods.

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Signature of Guide **Signature of Project Coordinators**